

Introduction to Microeconomics

Lecture 8—Consumer Theory: Preferences

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A model of consumer Preferences

- We start with a set of objects (bundles) X . Elements from from this set, like x or y , are the choices that the consumer can make.
- Most applications consider $X \subset \mathbb{R}_+^n$.
- In this class, $X = \mathbb{R}_+^2$. A bundle x is then $x = (x_1, x_2)$. Interpret it as: x_1 is the quantity of good 1 and x_2 is the quantity of good 2.

Consumer preferences: The Questionnaire Q

Consists of a long list of questions of the following form.

$Q(x, y)$, x and y are different bundles. How do you compare x to y ? Check one and only one of the following three options:

- ☐ I strictly prefer x to y ($x \succ y$).
- ☐ I strictly prefer y to x ($y \succ x$).
- ☐ I am indifferent between x and y ($x \sim y$).

A model of consumer preferences is just a long list of questions of this type and their answers!

What is a legal answer to the questionnaire?

- We need to put some restrictions (assumptions) on answers.

Restriction 1: Completeness

For any question $Q(x, y)$ exactly one of the boxes is checked. We don't allow

- Refraining to answer (no box checked).
- Checking more than one box.
- This means we exclude attitudes like:
 - They are not comparable.
 - I don't know what x is.
 - I have no opinion.
 - I prefer x over y and y over x .
- We are also not allowing attitudes like:
 - It depends on what others think.
 - It depends on the circumstances (frame).
 - Any consideration that includes **intensity**, like I somewhat prefer x , or I love x and hate y .

Restriction 2: Transitivity

Suppose I have bundles x, y, z such that $Q(x, y) : x \succ y$, and $Q(y, z) : y \succ z$. Then this implies that your answer to $Q(x, z)$ should be $x \succ z$.

Similarly, if $Q(x, y)$ is $x \sim y$ and $Q(y, z)$ is $y \sim z$ then in $Q(x, z)$ your answer should be $x \sim z$.

- Questionnaire Q could have been rephrased instead with the following type of questions:
 - $Q(x, y)$, x and y are different bundles. How do you compare x to y ? Check one and only one of the following three options:
 - ☐ I consider x at least as good as y ($x \succsim y$).
 - ☐ I consider y at least as good as x ($y \succsim x$).
- What is the relationship between the answers to questionnaire Q and the one above?
 - If $x \succsim y$ and $y \succsim x$ then $x \sim y$.
 - If $x \succsim y$ but $y \not\succsim x$ then $x \succ y$.
 - If $y \succsim x$ but $x \not\succsim y$ then $y \succ x$.

Suppose the set of bundles $X \subset \mathbb{R}_+^n$ consists of *goods that are desirable*.

Restriction 3: Monotonicity or “more-is-better”

If $x \geq y$, where $\geq$ denotes the component-wise standard order, then $x \succsim y$.

If $x > y$, where $>$ denotes the component-wise strict order, then $x \succ y$.

Example: say good 1 is food and good 2 is clothing. $x = (30, 40)$ and $y = (20, 30)$. If preferences satisfy monotonicity, then the answer to $Q(x, y)$ should be $x \succ y$.

- For a bundle x the **indifference curve at x** is the set of all bundles that are indifferent to x :

$$IC(x) = \{y \in X : y \sim x\}$$

- The strictly preferred set to x is the set of all bundles that are strictly preferred to x :

$$\succ(x) = \{y \in X : y \succ x\}$$

- The strictly worse set to x is the set of all bundles that are worse to x :

$$\prec(x) = \{y \in X : x \succ y\}$$

- Completeness implies that the indifference curve map is **full**, through each bundle there is an indifference curve passing through it.
- Transitivity implies that the indifference curves **cannot intersect!**
- Monotonicity implies that the indifference curves **are not thick!**